

Introduction

This report focuses on passive footprinting techniques used to gather publicly available information about a target domain. The objective is to understand how attackers can collect information without directly interacting with the system.

Target Selection

The target chosen for this footprinting activity is **unikl.edu.my**, which is the official website of Universiti Kuala Lumpur. This domain was selected because it is publicly accessible and suitable for passive reconnaissance techniques.

TITLE:

Footprinting vs Reconnaissance

Reconnaissance is the initial phase of ethical hacking where general information about a target is collected using passive methods such as search engines and social media. The goal is to understand the target broadly before focusing on specific systems.

Footprinting, on the other hand, is a more detailed process that involves gathering specific information such as domain details, IP addresses, and network structure. It may include both passive and light active techniques like WHOIS and DNS lookup.

Methodology

The methodology used in this assignment focuses on passive footprinting techniques (OSINT) to gather publicly available information about the target domain.

Several tools and techniques were used:

- WHOIS Lookup – to obtain domain registration details such as registrar, IP address, and nameservers
- DNS Lookup (nslookup / MXToolbox) – to identify IP addresses and mail servers
- Subdomain Enumeration – to discover related subdomains of the target
- Search Engine Footprinting (Google Dorking) – to find publicly available files and sensitive information
- Technology Identification (Wappalyzer) – to detect technologies used by the website

All information was collected without directly interacting with or attacking the target system.

WHOIS Lookup

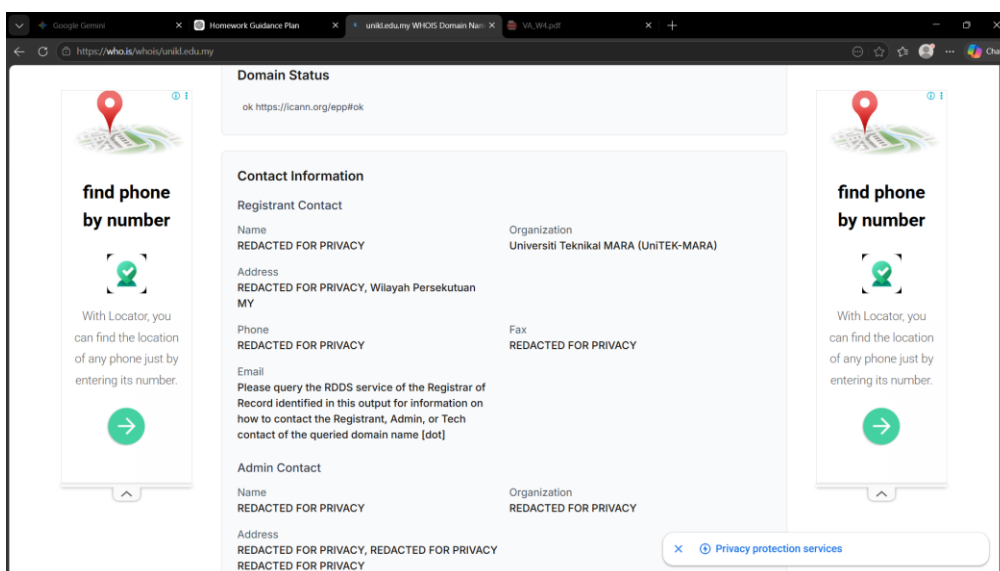
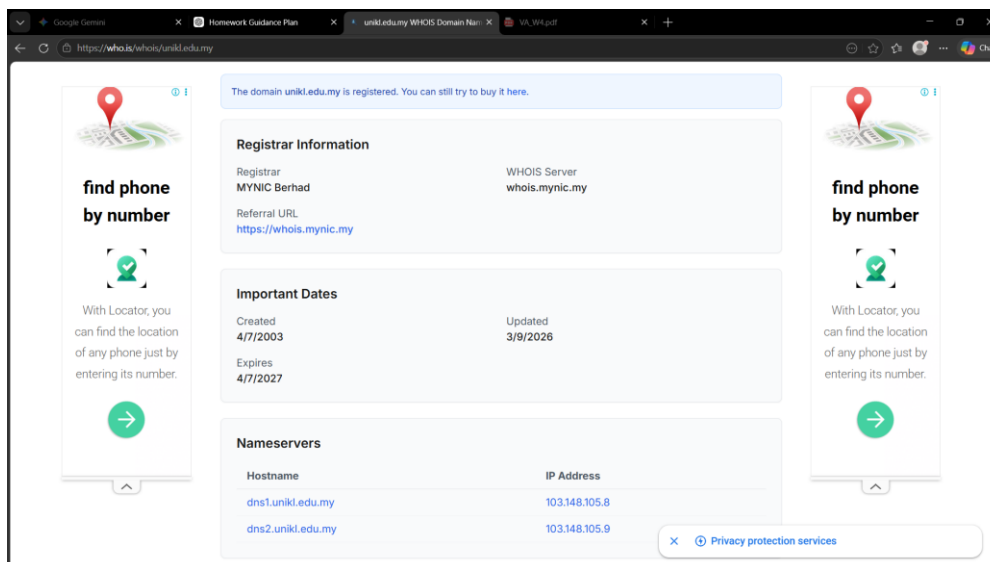
The WHOIS lookup was performed to gather domain registration details of the target domain.



Result

The WHOIS lookup for unkl.edu.my provided the following information:

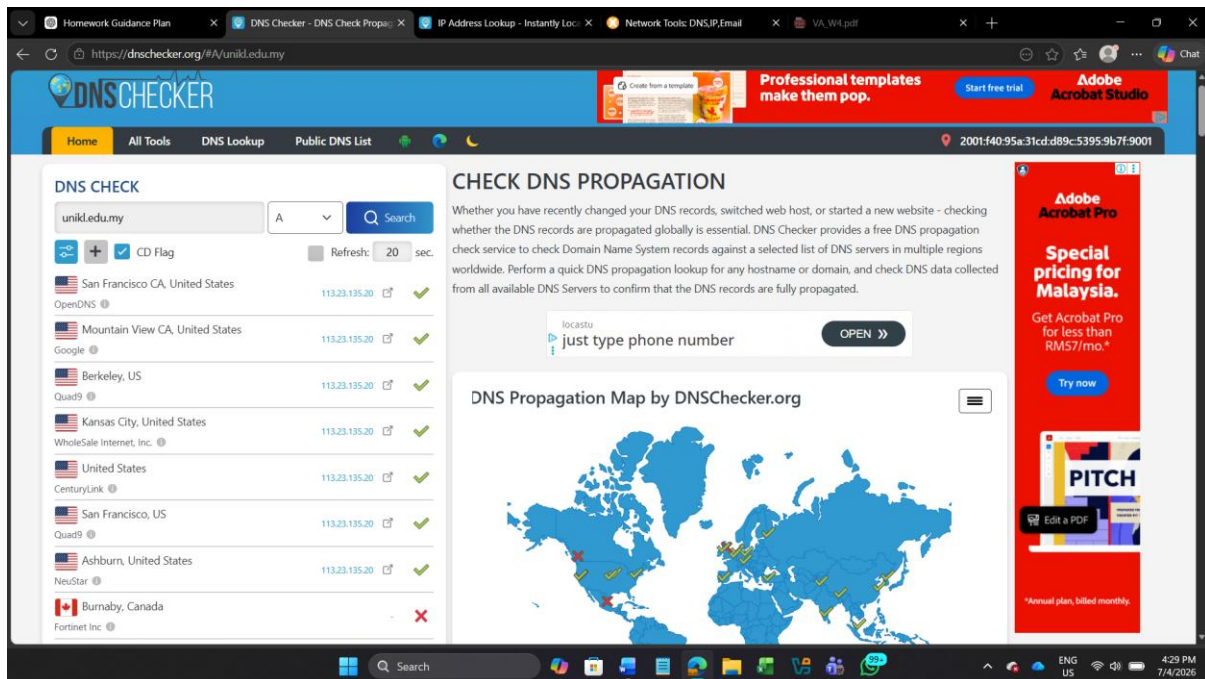
- Domain Name: unkl.edu.my
- IP Address: 113.23.135.20
- Registrar: MYNIC Berhad
- WHOIS Server: whois.mynic.my
- Creation Date: 4 July 2003
- Expiry Date: 4 July 2027
- Name Servers:
 - dns1.unkl.edu.my (103.148.105.8)
 - dns2.unkl.edu.my (103.148.105.9)



The WHOIS result shows that the domain unkl.edu.my is officially registered under MYNIC Berhad, which is the authorized registrar for Malaysian domains. The domain has been active since 2003, indicating it is a well-established and legitimate organization.

The IP address and nameserver information reveal part of the target's network infrastructure. This allows an attacker to identify hosting details and potential entry points for further reconnaissance. Additionally, the presence of internal nameservers suggests that the organization manages its own DNS services, which could be a potential target for misconfiguration or vulnerabilities.

DNS Lookup



The DNS lookup for **unkl.edu.my** provided the following information:

- **A Record (IP Address):** 113.23.135.20
- **MX Record (Mail Server):** unkl-edu-my.mail.protection.outlook.com
- **Mail Server IP Address:** 52.101.124.117

Analysis

The DNS lookup shows that the domain is hosted at IP address 113.23.135.20, which indicates the server location of the website.

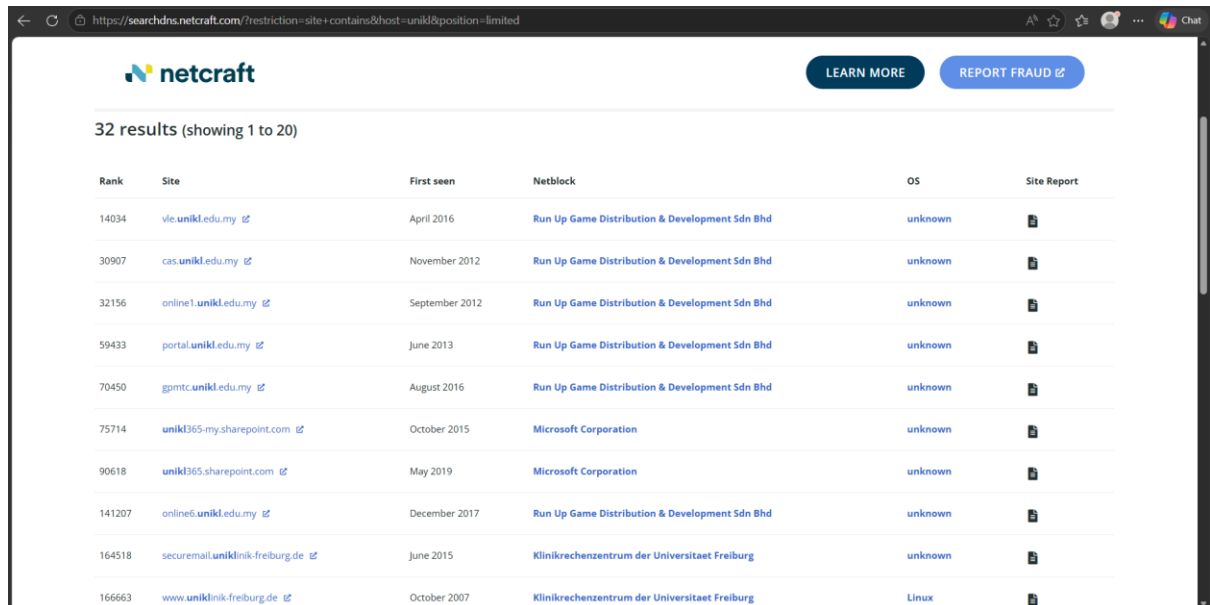
The MX record reveals that the domain uses Microsoft Outlook (Office 365) as its email service provider, indicating reliance on cloud-based infrastructure.

This information is important because DNS records expose critical system components such as web hosting and email services. Attackers can use this information to identify potential targets, such as mail servers for phishing attacks or system vulnerabilities for further exploitation.

SUBDOMAIN ENUMERATION

Description

Subdomain enumeration was performed to identify additional subdomains associated with the target domain unkl.edu.my.



The screenshot shows a web browser window with the URL <https://searchdns.netcraft.com/?restriction=site+contains&host=unkl&position=limited>. The page displays 32 results for the search criteria. The results are shown in a table with columns: Rank, Site, First seen, Netblock, OS, and Site Report. The first 10 results are visible, showing subdomains like vle.unkl.edu.my, cas.unkl.edu.my, online1.unkl.edu.my, portal.unkl.edu.my, gpmtc.unkl.edu.my, unkl365-my.sharepoint.com, unkl365.sharepoint.com, online6.unkl.edu.my, securemail.uniklinik-freiburg.de, and www.uniklinik-freiburg.de.

Rank	Site	First seen	Netblock	OS	Site Report
14034	vle.unkl.edu.my	April 2016	Run Up Game Distribution & Development Sdn Bhd	unknown	
30907	cas.unkl.edu.my	November 2012	Run Up Game Distribution & Development Sdn Bhd	unknown	
32156	online1.unkl.edu.my	September 2012	Run Up Game Distribution & Development Sdn Bhd	unknown	
59433	portal.unkl.edu.my	June 2013	Run Up Game Distribution & Development Sdn Bhd	unknown	
70450	gpmtc.unkl.edu.my	August 2016	Run Up Game Distribution & Development Sdn Bhd	unknown	
75714	unkl365-my.sharepoint.com	October 2015	Microsoft Corporation	unknown	
90618	unkl365.sharepoint.com	May 2019	Microsoft Corporation	unknown	
141207	online6.unkl.edu.my	December 2017	Run Up Game Distribution & Development Sdn Bhd	unknown	
164518	securemail.uniklinik-freiburg.de	June 2015	Klinikrechenzentrum der Universitaet Freiburg	unknown	
166663	www.uniklinik-freiburg.de	October 2007	Klinikrechenzentrum der Universitaet Freiburg	Linux	

RESULTS

The following subdomains were discovered:

- vle.unkl.edu.my
- cas.unkl.edu.my
- online1.unkl.edu.my
- portal.unkl.edu.my
- gpmtc.unkl.edu.my
- online6.unkl.edu.my

Analysis

The discovered subdomains indicate that the organization operates multiple services under the main domain, such as learning platforms (vle), authentication systems (cas), and internal portals.

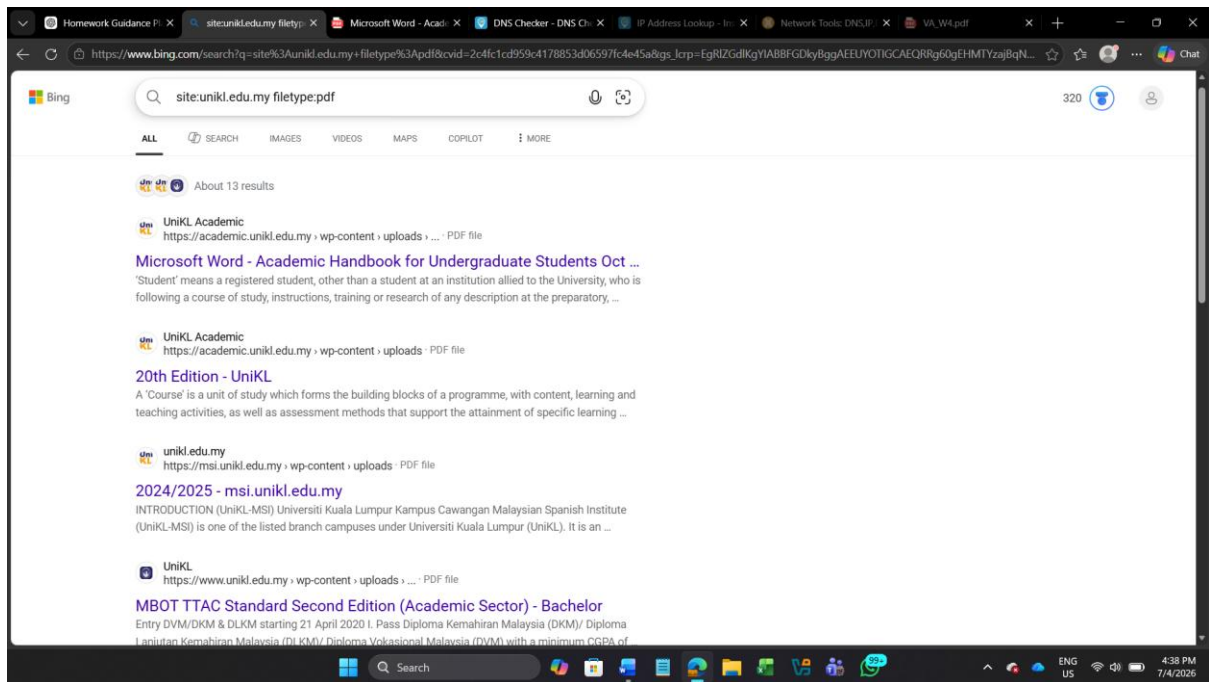
This information is important because each subdomain represents a potential entry point into the system. Attackers may target weaker or less-secured subdomains to gain access to the main network.

Subdomain enumeration helps reveal the structure of the organization's web infrastructure and identifies additional systems that may not be visible on the main website.

SEARCH ENGINE FOOTPRINTING (GOOGLE DORK)

Description

Search engine footprinting was performed using Google Dorking techniques to identify publicly accessible documents related to the target domain unkl.edu.my.



Results

Using the search query:

site:uninkl.edu.my filetype:pdf

Several publicly accessible PDF documents were discovered, including:

- Academic Handbook for Undergraduate Students
- UniKL academic documents and course information
- Institutional documents from different UniKL subdomains

Analysis

The results show that a large number of documents are publicly accessible through search engines. These files may contain useful information such as academic structure, internal processes, and organizational details.

This is important because attackers can use such documents to gather intelligence about the organization without directly interacting with its systems. In some cases, sensitive information may be unintentionally exposed, leading to potential information leakage.

Search engine footprinting highlights how publicly indexed data can become a security risk if not properly managed.

CONCLUSION

This assignment demonstrates how passive footprinting techniques can be used to gather valuable information about a target without direct interaction. Tools such as WHOIS, DNS lookup, subdomain enumeration, and search engine footprinting provide insights into the target's infrastructure, services, and publicly available data.

The findings show that a significant amount of information can be obtained from open sources, which may pose security risks if not properly managed. Therefore, organizations must be aware of information exposure and take appropriate measures to protect sensitive data.